

CLASS
6

QUESTION
PAPER SET

A



**SOF INTERNATIONAL
MATHEMATICS OLYMPIAD
2018-19**

DO NOT OPEN THIS BOOKLET UNTIL ASKED TO DO SO

Total Questions: 50 | Time: 1 hr.

Guidelines for the Candidate

1. You will get additional ten minutes to fill up information about yourself on the OMR Sheet, before the start of the exam.
2. Write your **Name, School Code, Class, Section, Roll No.** and **Mobile Number** clearly on the **OMR Sheet** and do not forget to sign it. We will share with you your marks / result on your mobile number.
3. The Question Paper comprises four sections:

Logical Reasoning (15 Questions), **Mathematical Reasoning** (20 Questions), **Everyday Mathematics** (10 Questions) and **Achievers Section** (5 Questions)

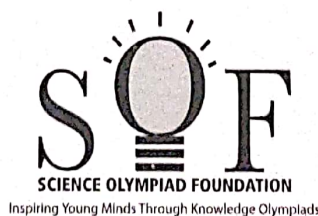
Each question in Achievers Section carries 3 marks, whereas all other questions carry one mark each.

4. All questions are compulsory. There is no negative marking. Use of calculator is not permitted.
5. There is only ONE correct answer. Choose only ONE option for an answer.
6. To mark your choice of answers by darkening the circles on the OMR Sheet, use **HB Pencil** or **Blue / Black ball point pen** only. E.g.
Q.16: Rahul bought 4 kg 90 g of apples, 2 kg 60 g of grapes and 5 kg 300 g of mangoes. The total weight of all the fruits he bought is _____.
A. 11.450 kg B. 11.000 kg C. 11.350 kg D. 11.250 kg

As the correct answer is option A, you must darken the circle corresponding to option A on the OMR Sheet.

16. ● (B) (C) (D)

7. Rough work should be done in the blank space provided in the booklet.
8. Return the OMR Sheet to the invigilator at the end of the exam.
9. Please fill in your personal details in the space provided on this page before attempting the paper.

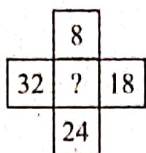
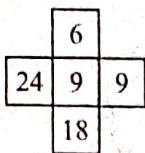
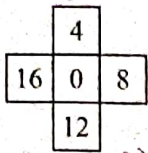


Name:.....

Section:..... SOF Olympiad Roll No.:..... Contact No.:.....

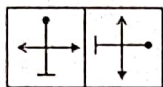


1. Find the missing number, if same rule is followed in all the three figures.



- A. 20
B. 4
C. 18
D. 25

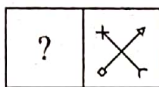
2. There is a certain relationship between figures (i) and (ii). Establish the similar relationship between figures (iii) and (iv) by selecting a suitable figure from the options which will replace the '?' in figure (iii).



(i)



(ii)



(iii)



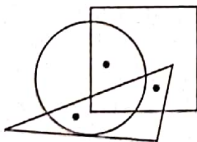
(iv)

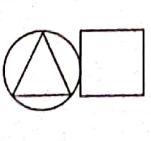
- A. 
B. 
C. 
D. 

3. If 'Racket' is called 'Football', 'Football' is called 'Cricket bat', 'Cricket bat' is called 'Basket ball' and 'Basket ball' is called 'Dice', then by which a cricketer play with?

- A. Cricket bat
B. Dice
C. Basket ball
D. Racket

4. Select a figure from the options which satisfy the same condition of placement of the dots as in the given figure.

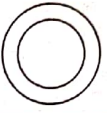


- A. 
B. 
C. 
D. 

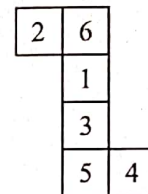
5. A, B, C and D are playing carrom. D is to the right of A who is to the right of B. Who is facing towards west, if B is facing towards North?

- A. D
B. C
C. A
D. Can't be determined

6. Which of the following Venn diagrams best represents the relationship amongst, 'Cats, Animals and Dogs'?

- A. 
B. 
C. 
D. 

7. The given sheet of a paper is folded to form a cube. Select a figure from the options which is similar to the cube formed by folding the given sheet.



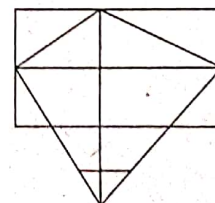
- A. 
B. 
C. 
D. 

8. How many 6's are there which are preceded by 9 but not followed by 3 in the given number series?

9 6 3 9 4 3 6 6 9 3 3 6 9 3 6 3 9 6 3 6 9

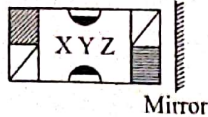
- A. 1
B. 2
C. 3
D. 0

9. Count the number of triangles formed in the given figure.



- A. 18
B. 15
C. 16
D. None of these

10. Select the correct mirror image of the given figure.

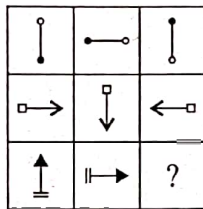


- A. B. C. D.

11. If '+' stands for subtraction, '-' stands for multiplication, '÷' stands for addition and '×' stands for division, then the value of $5 + 16 \times 4 \div 20 - 40$ is

- A. 799
B. 805
C. 801
D. 800

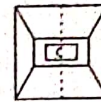
12. Select a figure from the options which will complete the given figure matrix.



- A. B. C. D.

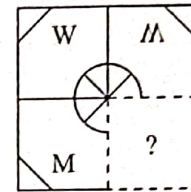
13. A square transparent sheet, with a pattern and a dotted line on it is shown here. Select a figure from

the options as to how the pattern would appear when the transparent sheet is folded along the dotted line.



- A. B. C. D.

14. Which of the following options would complete the given figure pattern?



- A. B. C. D.

15. Pointing to a lady in a photograph, Kashi said, "She is the mother of my son's wife's daughter". How is the lady related to Kashi?

- A. Daughter-in-law
B. Daughter
C. Granddaughter
D. Cousin

MATHEMATICAL REASONING

16. Find the difference between the smallest 5-digit number and the greatest 5-digit number formed by using the digits 6, 8, 2, 5 and 9 (using each digit only once).

- A. 27936
B. 72963
C. 56293
D. 39576

17. If the product of two numbers is 1728 and their H.C.F. is 12, then their L.C.M. is _____.

- A. 156
B. 144
C. 256
D. 172

18. Which of the following options gives the result CCC?

- A. $\text{MXIII} - \text{DCCXIV}$
B. $\text{LXIV} + \text{CCXXVI}$
C. $\text{CXXIX} + \text{CLXXI}$
D. $\text{XCVI} + \text{CCII}$

19. Which of the following properties holds true in the given expression?

$$72(4 + 5) = 72 \times 4 + 72 \times 5$$

- A. Commutative property
B. Closure property
C. Associative property
D. Distributive property

20. If a number is formed by interchanging the digits at tens and thousands places of 7939, then which of the following is CORRECT?

A. New number > Original number
 B. New number < Original number
 C. New number = Original number
 D. None of these

21. Find the minimum number of points at which three lines intersect.

A. 1
 B. 3
 C. 2
 D. 0

22. Using a ruler and compasses, which of the following constructions can be made?

A. A circle, when the length of its radius is known.
 B. The perpendicular bisector of a line segment of the given length.
 C. Both A and B
 D. None of these

23. On a number line, numbers are marked at a distance of 1 cm apart from each other. One end of a line segment is at -8 and the other end is at 9. How long is the line segment (in cm)?

A. 1
 B. -1
 C. 16
 D. 17

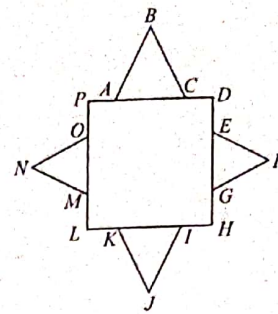
24. 1162 is divided into three parts such that 4 times the first part, 5 times the second part and 7 times the third part are equal. Find the parts.

A. 490, 392, 280
 B. 492, 392, 278
 C. 493, 329, 340
 D. 393, 290, 360

25. Find the fraction, if its decimal value is 0.6 and denominator is 75.

A. $\frac{35}{75}$
 B. $\frac{45}{75}$
 C. $\frac{65}{75}$
 D. $\frac{85}{75}$

26. In the given figure (not drawn to scale), $\triangle ABC$, $\triangle EFG$, $\triangle KIJ$ and $\triangle NMO$ are 4 different equilateral triangles. Find the perimeter of the figure, if side of square $PDHL$ is 8 cm, $AC = 4$ cm, $EG = 2$ cm, $KI = 3$ cm and $MO = 2$ cm.

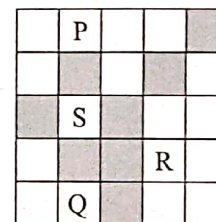


A. 34 cm
 B. 64 cm
 C. 43 cm
 D. 52 cm

27. If $a = 8$ and $x = 4$, then the value of $\frac{3ax + 6x - 9}{3a - 4x - 2}$ is _____.

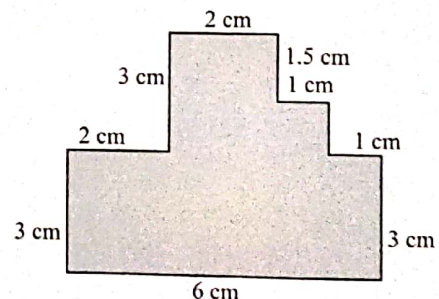
A. $18\frac{1}{3}$
 B. $19\frac{2}{7}$
 C. $19\frac{1}{3}$
 D. $18\frac{1}{2}$

28. Which of the following squares should be shaded to make the given figure symmetric?



A. P and Q
 B. P and S
 C. Q and R
 D. R and S

29. Find the area of the given figure.



A. 25.5 sq. cm
 B. 23.5 sq. cm
 C. 25 sq. cm
 D. 24 sq. cm

30. Read the statements carefully and select the CORRECT option.

Statement-I : A circle is the path of a point moving at the same distance from a fixed point.

Statement-II : A sector of a circle is a region in the interior of the circle enclosed by an arc and a chord.

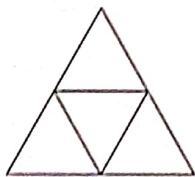
- A. Both Statement-I and Statement-II are true.
 B. Statement-I is true but Statement-II is false.
 C. Statement-I is false but Statement-II is true.
 D. Both Statement-I and Statement-II are false.

31. Arrange the following in ascending order.

$$\frac{6}{15}, \frac{2}{9}, \frac{4}{7}, \frac{3}{18}$$

- A. $\frac{4}{7}, \frac{2}{9}, \frac{6}{15}, \frac{3}{18}$
 B. $\frac{3}{18}, \frac{6}{15}, \frac{2}{9}, \frac{4}{7}$
 C. $\frac{3}{18}, \frac{2}{9}, \frac{6}{15}, \frac{4}{7}$
 D. $\frac{2}{9}, \frac{3}{18}, \frac{4}{7}, \frac{6}{15}$

32. The net of a solid is given. Identify the shape and find the number of faces of the solid formed when the net is folded.

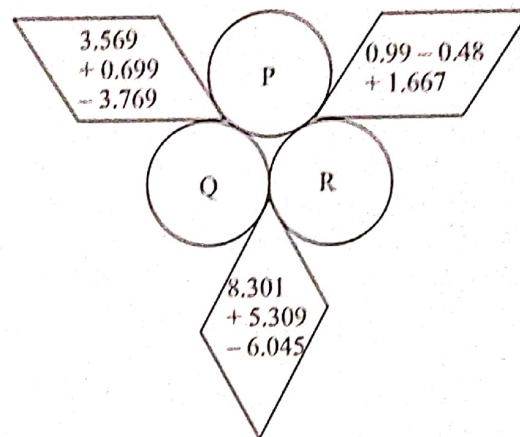


- A. Triangular pyramid, 4
 B. Square pyramid, 5
 C. Triangular prism, 5
 D. Tetrahedron, 6

33. Which of the following numbers is completely divisible by 9?

- A. 596348
 B. 965864
 C. 695844
 D. 746936

34. Find the values of circles P, Q and R respectively, such that the sum of the numbers in two quadrilaterals equals to the value of the circle in between them.



- A. 2.676, 8.064, 9.742
 B. 7.334, 6.442, 2.332
 C. 9.742, 2.676, 8.064
 D. 8.064, 9.742, 8.999

35. Which of the following ratios are in proportion?

- A. 150 g : 350 g and 35 kg : 210 kg
 B. ₹60 : ₹120 and ₹90 : ₹160
 C. 24 cm : 15 m and 80 g : 5 kg
 D. 3 L : 9 L and 18 mL : 27 L

EVERYDAY MATHEMATICS

36. A retailer bought 12000 strawberries. He threw 144 strawberries that were rotten. He packed the remaining strawberries equally in 76 boxes. How many strawberries were there in each box?

- A. 156
 B. 149
 C. 160
 D. 250

37. Ruhanika is organising a party for 480 people and needs disposable glasses and straws. There are 40 glasses in a pack and 160 straws in a pack. She needs exactly the same number of glasses and straws. What

is the minimum number of each pack she must buy?

- A. 14 packs of glasses and 12 packs of straws
 B. 15 packs of glasses and 3 packs of straws
 C. 14 packs of glasses and 11 packs of straws
 D. 12 packs of glasses and 3 packs of straws

38. Five square flower beds each of side 1.2 m are dug on a piece of land 4.8 m long and 4.2 m wide. What is the area of the remaining part of land?

- A. 12.69 sq. m
 B. 12.96 sq. m
 C. 11.96 sq. m
 D. 144 sq. m

39. At present, Kirti is 12 years old and her mother Kamlesh is 42 years old. Find the ratio of the Kamlesh's age two years ago to Kirti's age two years hence.

A. 7 : 20
B. 20 : 7
C. 10 : 3
D. 3 : 10

40. The breadth of a rectangular bed sheet is 5 m more than half the length of the bed sheet. What is the perimeter of the bed sheet, if the length is x m?

A. $(3x + 12)$ m
B. $2(x + 5)$ m
C. $(3x + 10)$ m
D. $(4x + 12)$ m

41. On one day, the temperature on a hill at 8 p.m. was 2°C but at mid-night, it fell down to -3°C . By how much did the temperature fall?

A. 5°C
B. 6°C
C. 7°C
D. 8°C

42. Chinmay purchased 10 kg rice at the rate of ₹ 15 per kg, 15 kg 40 g sugar at the rate of ₹ 20 per kg and 5 kg 60 g salt at the rate of ₹ 9 per kg. He gave ₹ 1000 to the shopkeeper. How much money did he get back?

A. ₹ 234.94
B. ₹ 503.66

C. ₹ 496.34
D. ₹ 743.99

43. Swati ate $\frac{1}{2}$ of the candies she had and gave rest to Jeny. Jeny kept 8 of the candies and gave rest of the 10 candies to Sakshi. How many candies did Swati eat?

A. 18
B. 24
C. 72
D. 36

44. Three farmers have 165 kg, 190 kg and 210 kg of wheat respectively. Find the maximum capacity of a bag such that the quantity of wheat can be packed in exact number of bags.

A. 15 kg
B. 50 kg
C. 45 kg
D. 5 kg

45. Vishu writes the smallest 6-digit number and Ridhi writes the greatest 7-digit number on the blackboard. Their Maths teacher finds the difference between the smallest 4-digit number and the sum of the two numbers written on the board. What would be the answer?

A. 10099999
B. 10098999
C. 19999890
D. 10069998

ACHIEVERS SECTION

46. Fill in the blanks.

- (i) A right angle is P of a revolution.
(ii) A figure whose all sides are equal and all angles are Q is called regular closed figure.
(iii) The product of two negative integers is a R integer.
(iv) S of a number is a number that comes just after that number.

	P	Q	R	S
A.	One-fourth	different	positive	successor
B.	One-third	different	negative	predecessor
C.	One-fourth	equal	positive	successor
D.	Half	equal	negative	predecessor

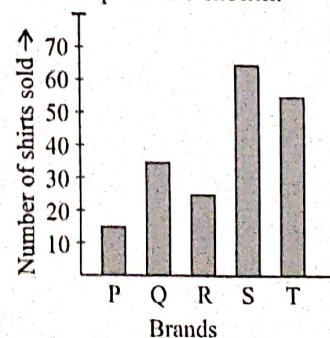
47. State 'T' for true and 'F' for false and select the correct option.

- (i) Set-squares are useful to draw parallel lines.
(ii) $13.5 + 4\frac{3}{8} + 9\frac{1}{8} + 6\frac{4}{8} > 12.5 + 5\frac{1}{2} + 8 + 14.6$

- (iii) Zero is less than every negative number.
(iv) 4 more than -7 is -3 .

	(i)	(ii)	(iii)	(iv)
A.	T	T	F	T
B.	T	F	F	T
C.	T	T	F	F
D.	F	T	F	T

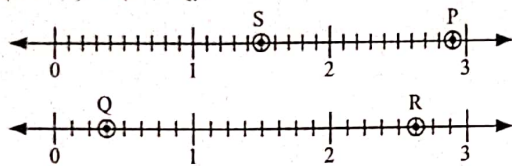
48. The given bar graph shows the sale of different brands of shirts in a shop in one month.



- (i) How many total shirts are sold in the month?
 (ii) Find the fraction of number of shirts sold of brand S to the number of shirts sold of brand P and R together.

(i)	(ii)
A. 195	$\frac{8}{9}$
B. 200	$\frac{13}{8}$
C. 165	$\frac{11}{7}$
D. 195	$\frac{13}{8}$

49. Study the given number lines and find the value of $(S + R) \div (P - Q)$.



- A. $1\frac{64}{101}$
 B. $1\frac{512}{101}$
 C. $2\frac{512}{403}$

D. $1\frac{61}{213}$

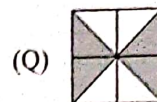
50. Match the figures given in Column-I with their shaded ratio given in Column-II and select the correct option.

Column-I

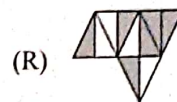
Column-II



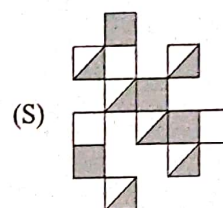
(i) 3 : 5



(ii) 7 : 13



(iii) 7 : 12

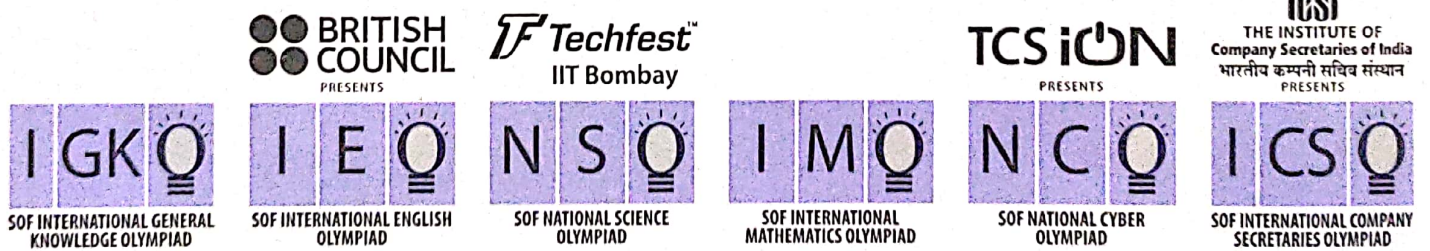


(iv) 1 : 2

- A. (P) → (iii), (Q) → (iv), (R) → (i), (S) → (ii)
 B. (P) → (iv), (Q) → (iii), (R) → (ii), (S) → (i)
 C. (P) → (iv), (Q) → (iv), (R) → (i), (S) → (ii)
 D. (P) → (iii), (Q) → (ii), (R) → (iv), (S) → (i)

SPACE FOR ROUGH WORK

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